

Neural Networks for Safe Robotics Control

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Abstract

In this project, we will use MuJoCo to simulate a quadcopter drone and develop a neural network controller for it. The main task will be to make the quadcopter reach a target position or follow a set of waypoints in a human environment, where the human action will be random. We will first create a basic controller and then train a neural network controller. We will test both controllers and compare how well they perform. And we will have two layers of controller, where one controller will work when there is no disturbance and another controller will work when it detects disturbance. If we have enough time, we will also explore using a similar approach to a simulated humanoid robot.

Introduction

Neural networks can be used to help robots learn how to move and complete tasks. Using simulation makes it easier and safer to test different controls before using a real robot.

For this project, we will use MuJoCo to simulate a quadcopter drone. We will first learn how to control and move the quadcopter in the simulation. We will then create a simple navigation task, such as reaching a target position or following waypoints.

Our main goal is to develop a neural network controller for the quadcopter and compare it with a basic controller. We will also test how the controller performs when small changes or disturbances are added to the simulation. We want to use two sets of controllers, which are inspired by humans, as we also go in and out of zone and use more brain power for some task when there is disturbance, but if there is a clear way to do it, we just do it. It was also inspired while reading paper from Heydari Shahna and Matilla [3], but it may all change during our research, but for now that is the idea.

Literature Review

[1] Binny et al. develop a neural-network-based approach for learning robot motion while also considering safety and stability. Their method learns from demonstrations and uses safety and stability measures to help the robot reach its goal while avoiding unsafe regions. This work motivates us to consider both task performance and safety when testing our neural-network controller.

[2] Zhang et al. review different approaches for safe learning in robotics, including safe exploration, safe execution, safety filters, and methods for handling uncertainty. They also discuss the importance of simulation for safely testing learning-based controllers and consider safety in human-robot environments. These ideas are useful for our project because we plan to test the quadcopter under disturbances and around a simulated human.

[3] Heydari Shahna and Mattila combine a neural-network controller with a robust adaptive controller for wheeled mobile robots. The neural-network controller is used during normal operating conditions, while the robust controller takes over when disturbances cause the system performance to cross a predefined limit. This layered approach directly inspired our idea of using different control responses during normal and disturbed conditions. Their method was developed for wheeled robots, so we will explore whether a similar idea can be applied to a simulated quadcopter.

Methods and Schedule

We will begin by setting up MuJoCo and loading a quadcopter model. Our first step will be to understand how the quadcopter moves and how its motors can be controlled in the simulation.

After basic control is working, we will create a simple environment and give the quadcopter a task. The task will focus on moving from a starting position to a target or following several waypoints.

We will first create a basic controller so that we have something to compare against. We will then develop a neural network that takes information about the quadcopter, such as its position, movement, and target position, and produces the controls needed to move the drone.

After training the neural network, we will run multiple tests and compare its performance with the basic controller. We will also introduce small changes or disturbances in the simulation to see how well the controller handles different situations.

Tools

- **MuJoCo** – for running the robot simulation.
- **Python** – for programming and controlling the simulation.
- **PyTorch or TensorFlow** – for developing and training the neural network.
- **Git/GitHub** – for managing and sharing our code.

We plan to run the simulation and train the neural network on our computers. We will keep the initial simulation and neural network simple so that we do not need large computing resources.

Weekly Schedule

Week	Work
Week 1	Set up MuJoCo and learn the basic simulation environment.
Week 2	Load the quadcopter model and test basic controls.
Week 3	Create a target or waypoint task for the quadcopter.
Week 4	Develop and test a basic controller.
Week 5	Start developing the neural-network controller.
Week 6	Train the neural-network controller.
Week 7	Test the controller and make improvements if needed.
Week 8	Compare the neural-network controller with the basic controller.
Week 9	Test with small changes or disturbances in the simulation.
Week 10	Collect and compare the results.
Week 11	Explore the humanoid simulation if time allows.
Week 12	Complete testing, demonstration, and the final report.

Anticipated Results

By the end of the project, we expect to have a working quadcopter simulation in MuJoCo with a neural-network controller that can complete a basic navigation task.

We will test how well the quadcopter reaches its target and compare the neural-network controller with the basic controller. We can compare things such as the distance from the target, time taken to complete the task, and whether the quadcopter successfully completes the task.

We also plan to create a simple graph to compare the two controllers. For example, the X-axis can show each test run and the Y-axis can show the distance from the target. This will help us clearly see how the two controllers perform.

If time allows, we will also explore using a simulated humanoid robot.

Technical Resources

The following starter references were provided for the project:

1. CrazySim – Crazyflie software-in-the-loop simulation
<https://github.com/gtfactslab/CrazySim>
2. MuJoCo – Robotic simulation and drone models
<https://mujoco.readthedocs.io/en/3.3.3/models.html#drones>

References

- [1] A. E. Binny et al., “Safe and Stable Neural Network Dynamical Systems for Robot Motion Planning,” arXiv:2511.20593, 2025. <https://arxiv.org/abs/2511.20593>
- [2] H. Zhang et al., “Safe Learning for Contact-Rich Robot Tasks: A Survey from Classical Learning-Based Methods to Safe Foundation Models,” arXiv:2512.11908, v2, 2026. <https://arxiv.org/abs/2512.11908>
- [3] M. Heydari Shahna and J. Mattila, “Synthesis of Deep Neural Networks with Safe Robust Adaptive Control for Reliable Operation of Wheeled Mobile Robots,” IEEE Transactions on Automation Science and Engineering, vol. 23, pp. 7456–7473, 2026. <https://doi.org/10.1109/TASE.2026.3679545>

AI Assistance

The AI tool Open AI Codex was used to help find relevant research papers, understand some of the concepts related to the project, and improve the writing and organization of the proposal. The research sources and AI suggestions were reviewed by the team before being included in the proposal.